



Using Health Sentinel

Log a meal, connect simulated MCPs, get a guardrailed daily report

A practical walkthrough: what to grant consent to, what a meal/lab-report/SMS run looks like, how to read the risk dashboard and the verifier-vs-critic panel, and how the nutritionist approval step works.

Streamlit UI

Meal + lab report input

SMS checkbox review

Human-approved release

WHO THIS IS FOR



End users

Log a meal, review SMS-derived signals, and read a daily report grounded in your own history and documents.



Nutritionist reviewers

Approve, modify, or reject predictions flagged for review, within a severity-based SLA.



Support / on-call

Investigate anomalies and guardrail breaches via the traceability report and typed fault classes.



Data scientists

Review the deterministic bias-variance check output for cohort-level accuracy drift.

GETTING STARTED

1

Grant Tier-1 consent (image/text processing + "not medical advice" acknowledgement) — nothing runs without it.

2

Pick a seeded profile, describe or photograph a meal, and optionally upload a lab report.

3

Review the SMS-derived checkboxes (meal timing, gym subscription, health messages) and uncheck anything not applicable.

SEEDED DEMO PROFILES

PROFILE	WHAT IT EXERCISES
demo-user	Prediabetes + peanut allergy — allergen filtering, glucose-trend severity bump
healthy-baseline-user	No conditions/allergies — tests against over-flagging
hypertension-user	Hypertension + shellfish allergy — sodium flag path + a second allergen
family-history-user	Family history of heart condition — exercises blocked-category suppression against real medical-RAG evidence

READING A RESULT



Deterministic verifier vs. LLM critic

The verifier (pure Python) is authoritative and shown first; the critic is an advisory second opinion that never overrides it — both panels are shown side by side.



Nutritionist review gate

Low confidence, a new user, HIGH/CRITICAL severity, or a verifier failure all pause the run. Approve, modify, or reject — nothing is shown to the user without a decision.

LIMITS, HONESTLY STATED

What it won't do. This is **not** a medical diagnosis tool — predictions are lifestyle patterns, never a disease claim. MCP connectors (iWatch, SMS, calendar, Google Drive) are all **simulated** with deterministic synthetic data; no real OAuth calls are made. The checkpointner, store, and rate limiter are in-memory and reset on process restart — only the Chroma index and the SQLite metrics store survive a restart.

BEST PRACTICES

- Upload a real lab report (or a plain-text one) rather than typing numbers manually — extraction is validated against physiological bounds before anything is stored.
- Review the SMS checkboxes every run — keyword matching is approximate by design; your confirmation is the safety net.
- Treat a "degraded" run status as informative, not alarming — it means an agent recovered from a fault, and the recovered-fault count is shown.
- Use the sidebar's "Generate traceability report" button to refresh the audit trail on demand instead of looking for a live table in the app.

Get a grounded answer, not a guess.

Every prediction traces back to your profile, documents, or history — reviewed by a deterministic verifier before you see it.

Streamlit UI
localhost:8501

Grounded · Verified · Yours